Unmasking the Ring: Scalable Fraud Detection at the 6M Record Scale

A Multi-Stage Batch Processing & Graph Analysis Approach

$4 Billion in High-Risk Volume Identified

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Project Journey (The "How" and "Why")

The Ingestion Phase:

- Decision: Load the PaySim dataset into Neo4j. Indexed nodes. Eliminated duplicates.

The Detection Phase:

- Ran Cypher queries and GDS algorithms against the graph to flag fraudulent accounts, transactions, and rings. Wrote findings back onto the graph as node and relationship properties (risk scores, ring ids, flags, community assignments).
- Cycles were not identified in the data.

Challenge: Hardware limitations were causing errors.

- Solution: Optimized queries by creating indices and adjusted the maximum Java heap size to address out of memory errors

Detection Findings (The "Signals")

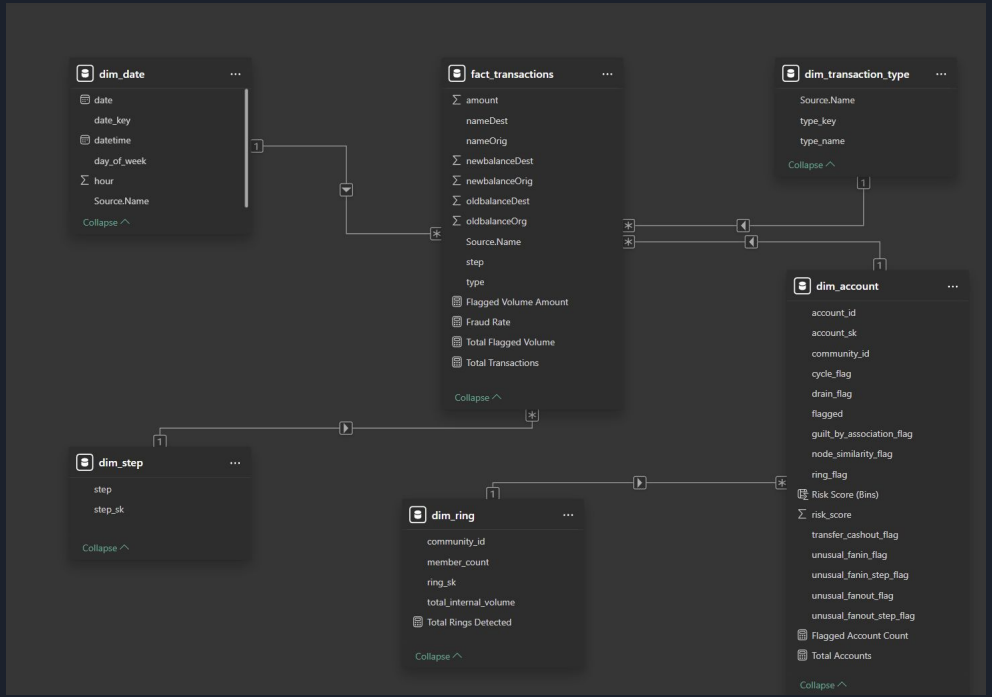
Feature Engineering: Implemented signals such as:

1. Fan-In: Identified accounts receiving money from an unusual number of sources.
2. Dense community with high-volume: found clusters of accounts moving lots of money among its own members.
3. Guilty-by-association: Accounts that transact frequently with flagged accounts but haven't been flagged themselves.

The Risk Score: Combined signals into a weighted score (0–100).

- Weighting Justification: Community/Ring membership was weighted highest (40%), as organized fraud is a higher threat than individual high-volume transfers.

The Star Schema



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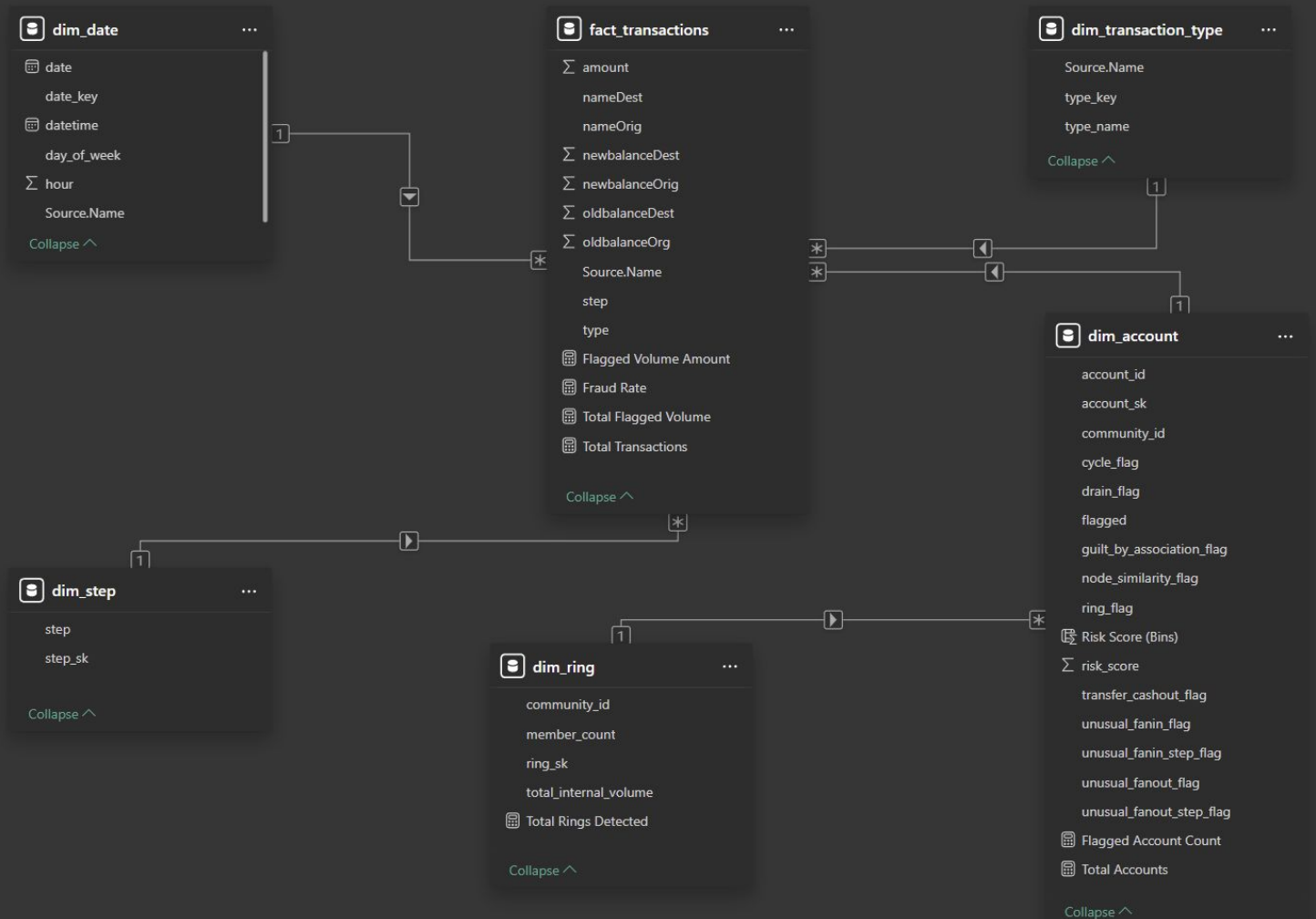
dim_date
date
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Source.Name
Collapse
fact_transactions
amount
nameDest
nameOrig
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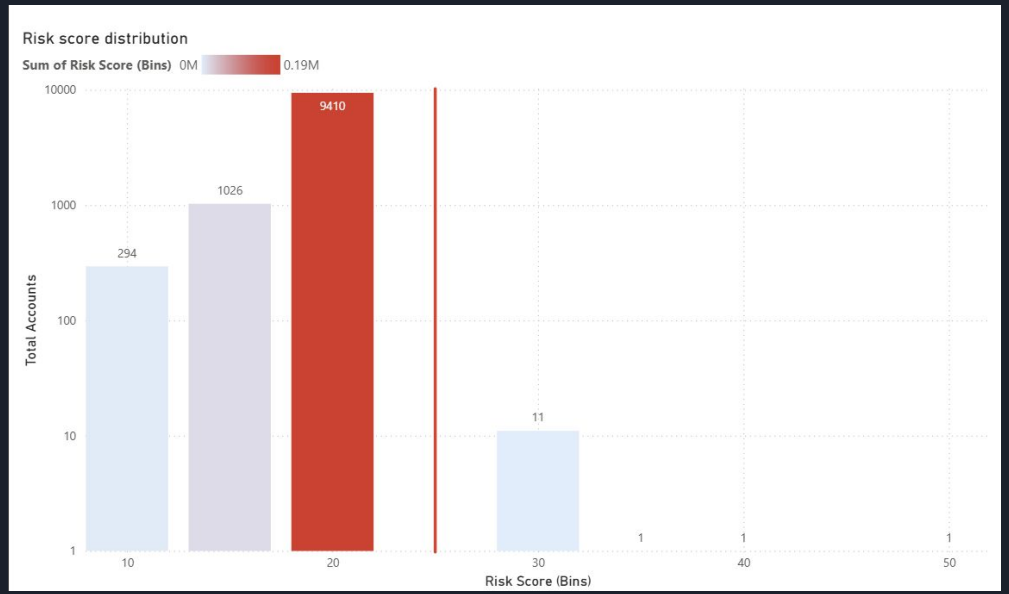
newbalanceOrig
oldbalanceDest
oldbalanceOrg
Source.Name
step
type
Flagged Volume Amount
Fraud Rate
Total Flagged Volume
Total Transactions
Collapse
dim_transaction_type
Source.Name
type_key
type_name
Collapse
dim_account
account_id
account_sk
community_id
cycle_flag
drain_flag
flagged
guilt_by_association_flag
node_similarity_flag
ring_flag
Risk Score (Bins)
risk_score
transfer_cashout_flag
unusual_fanin_flag
unusual_fanin_step_flag
unusual_fanout_flag
unusual_fanout_step_flag
Flagged Account Count
Total Accounts
Collapse
dim_step
step
step_sk
Collapse
dim_ring
community_id
member_count
ring_sk
total_internal_volume
Total Rings Detected
Collapse
1
*

Risk Score Distribution (The "Defense")



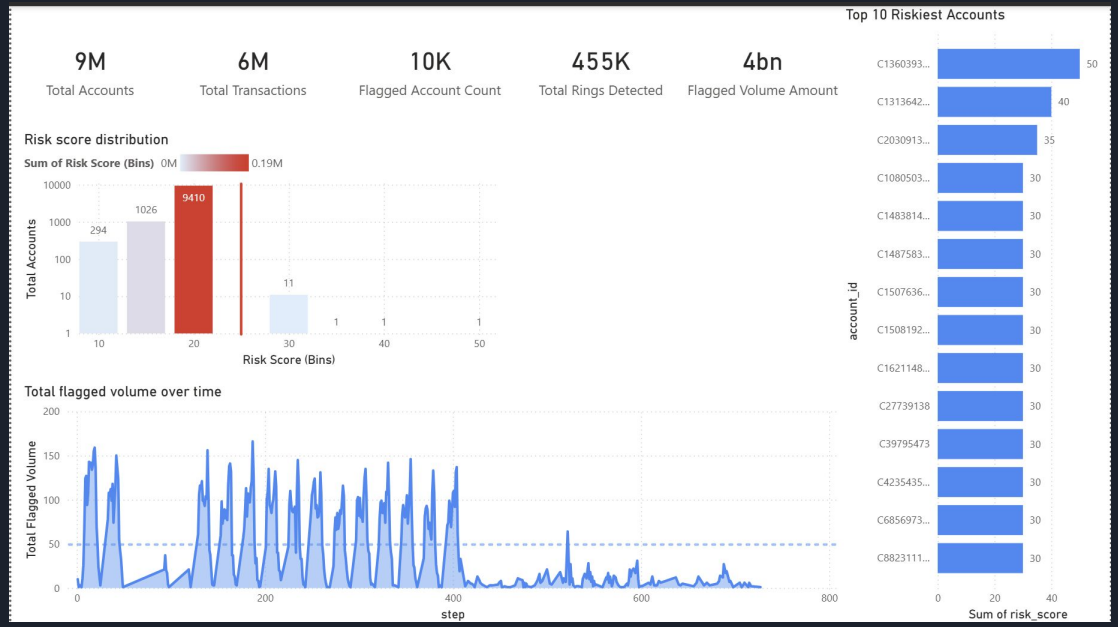
The "Natural Break": Show the "Valley" between scores 20 and 30.

The Threshold: "We set our threshold at 25 because it separates the 'noisy' population (thousands of minor flags) from the 'anomalous' population (14 high-intensity accounts)."

Dashboard Walkthrough (Part 1: Executive Summary)

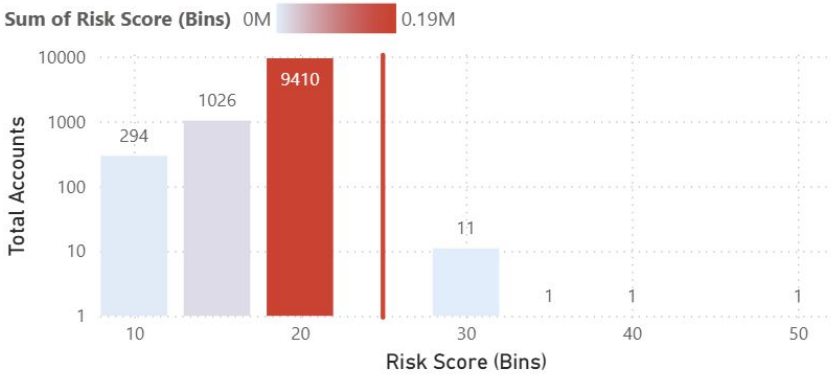
Key Findings:

- $4bn in flagged volume across only 14 accounts.
- Spikes represent coordinated attack dates.

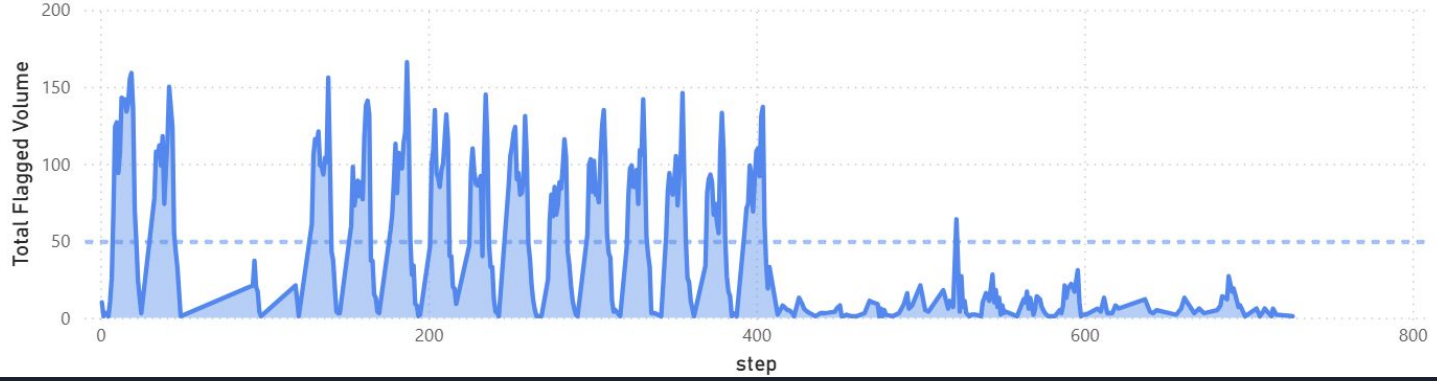


9M	6M	10K	455K	4bn
Total Accounts	Total Transactions	Flagged Account Count	Total Rings Detected	Flagged Volume Amount

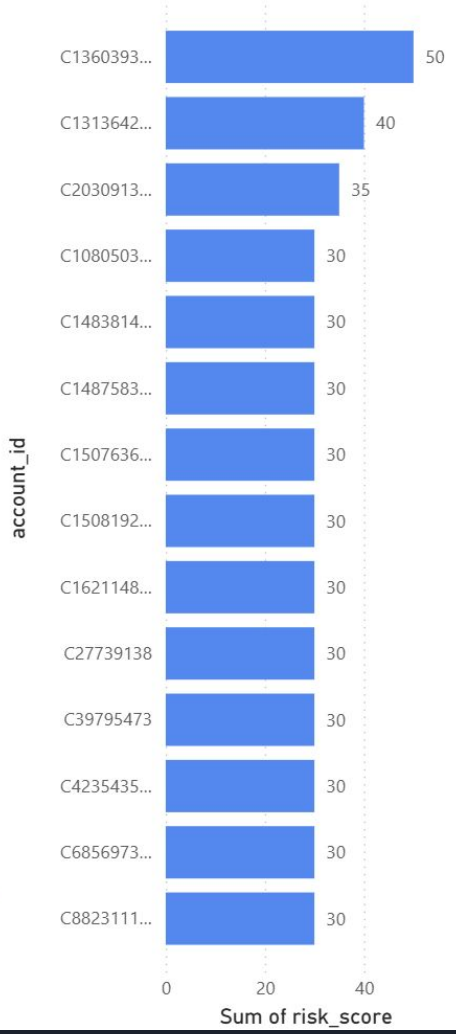
Risk score distribution



Total flagged volume over time



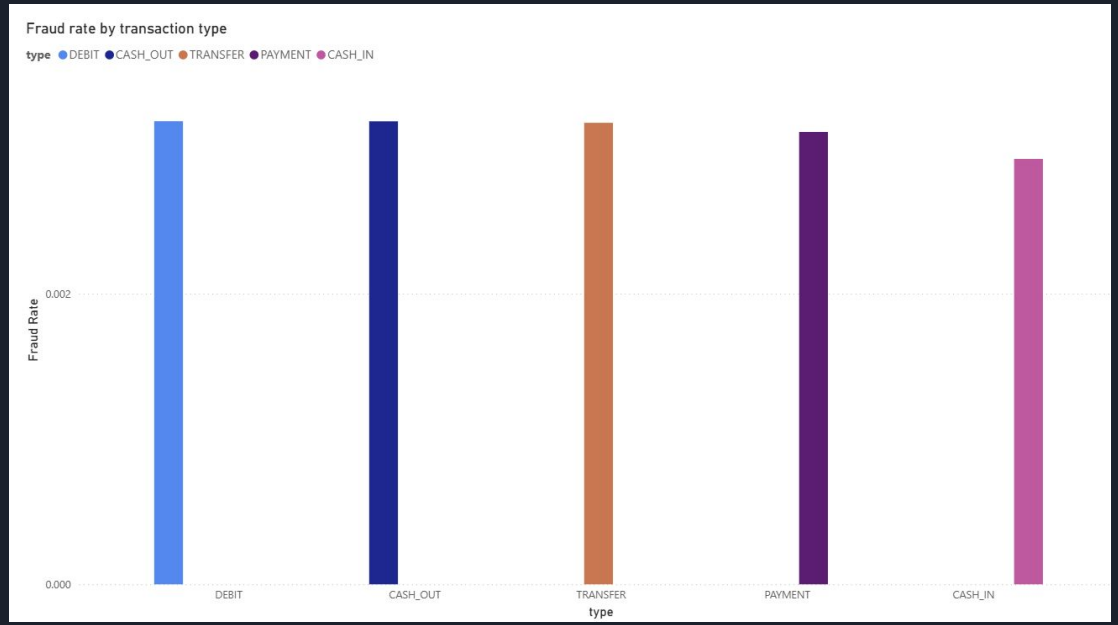
Top 10 Riskiest Accounts

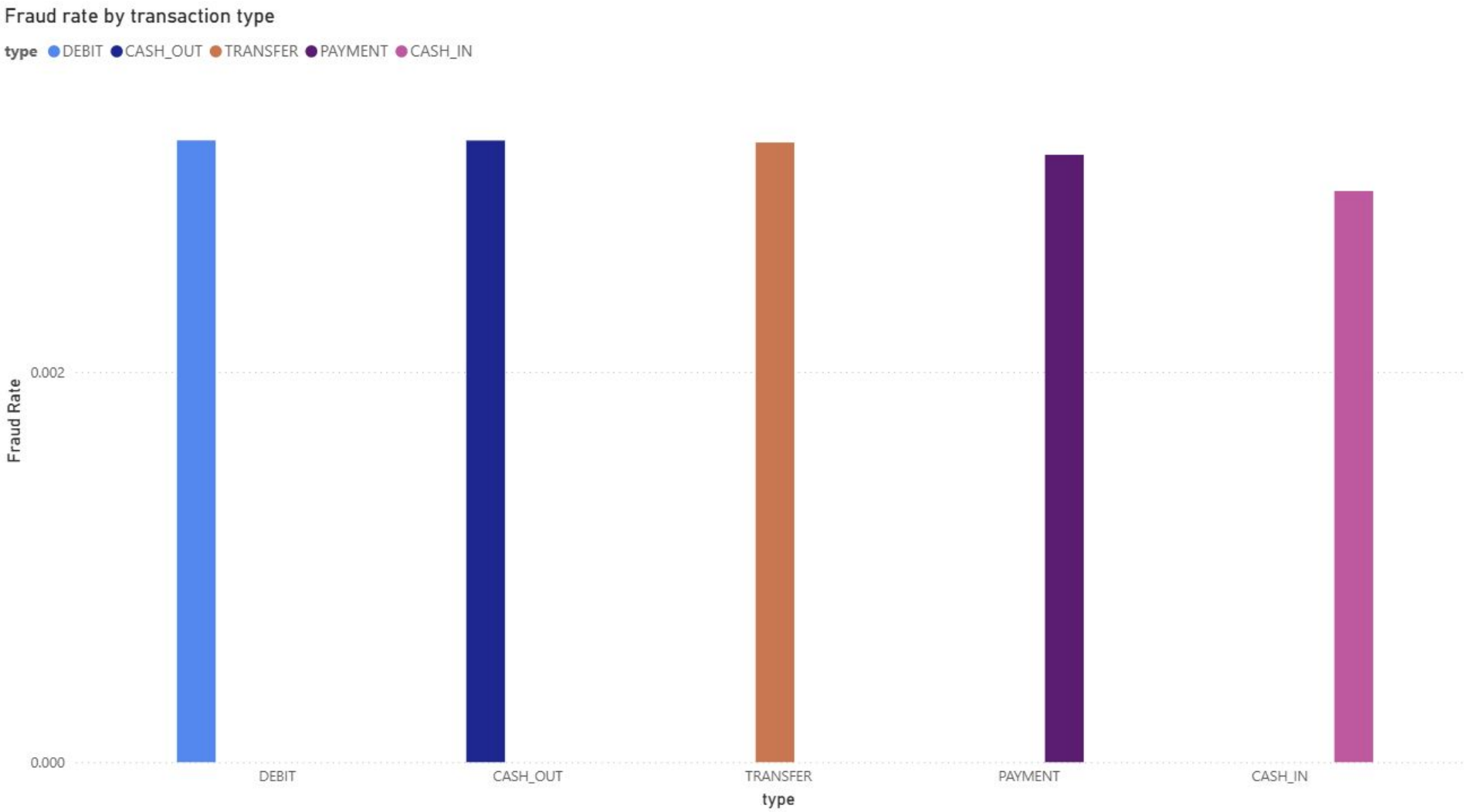


Dashboard Walkthrough (Part 2: Transaction Types)

The Insight: TRANSFER and CASH_OUT show significantly higher fraud rates than PAYMENT.

Unexpected Reveal: Despite PAYMENT having high transaction volume, its fraud rate was near zero, validating our focus on peer-to-peer transfers.

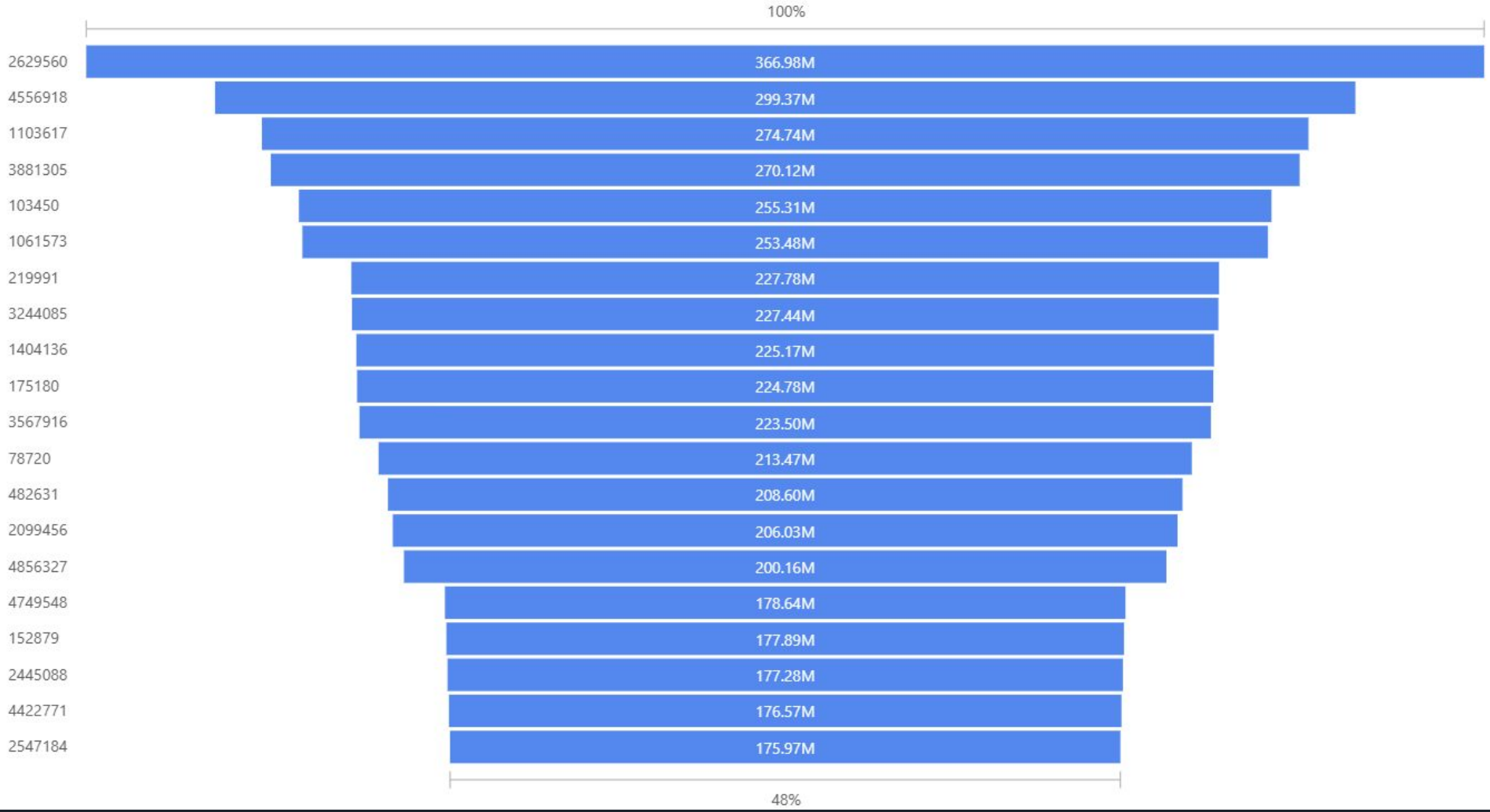



Fraud rate by transaction type
type DEBIT CASH_OUT TRANSFER PAYMENT CASH_IN
Fraud Rate
0.002
0.000
DEBIT
CASH_OUT
TRANSFER
PAYMENT
CASH_IN
type

Network Analysis (The "Rings")

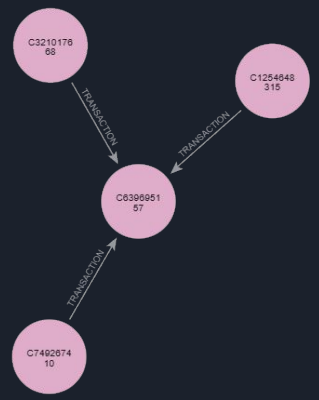
The Finding: Identified rings which moved massive amounts of money.

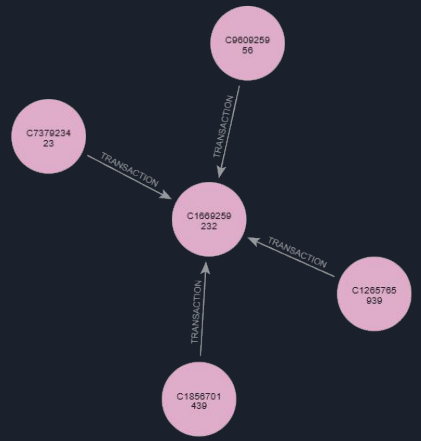
Which rings moved the most money

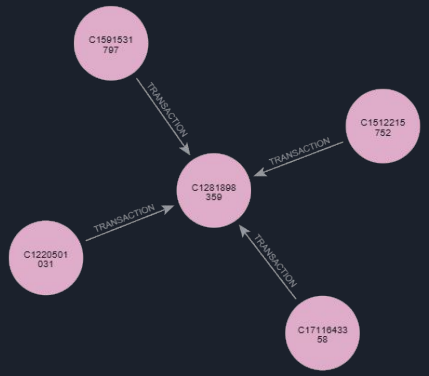


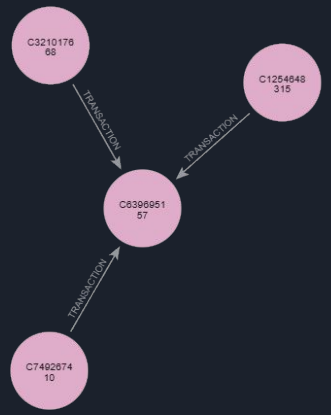
Case Study #2: Guilt by association cluster

- Shows an unflagged account connected to multiple flagged neighbors, revealing the social network around known bad actors





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C7492674 10
TRANSACTION
TRANSACTION
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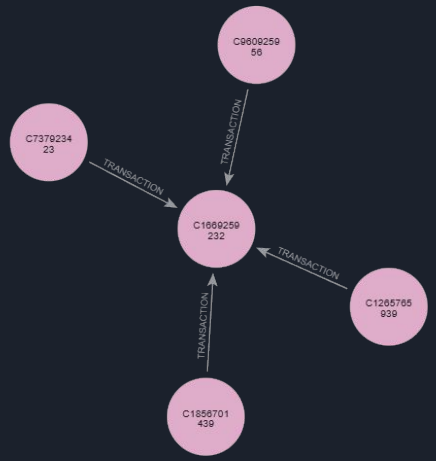
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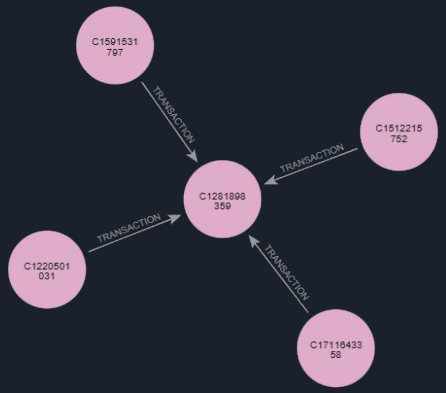
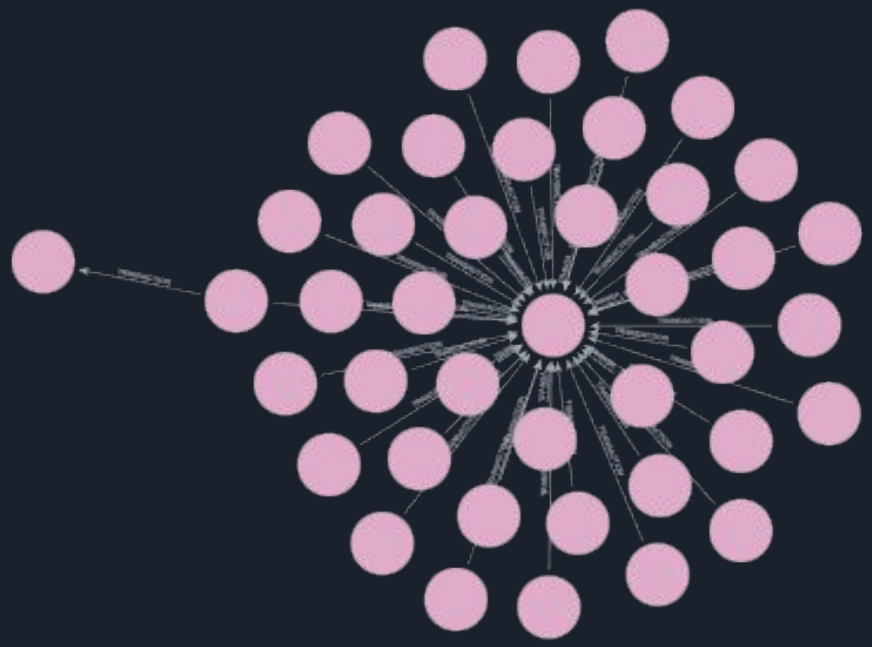
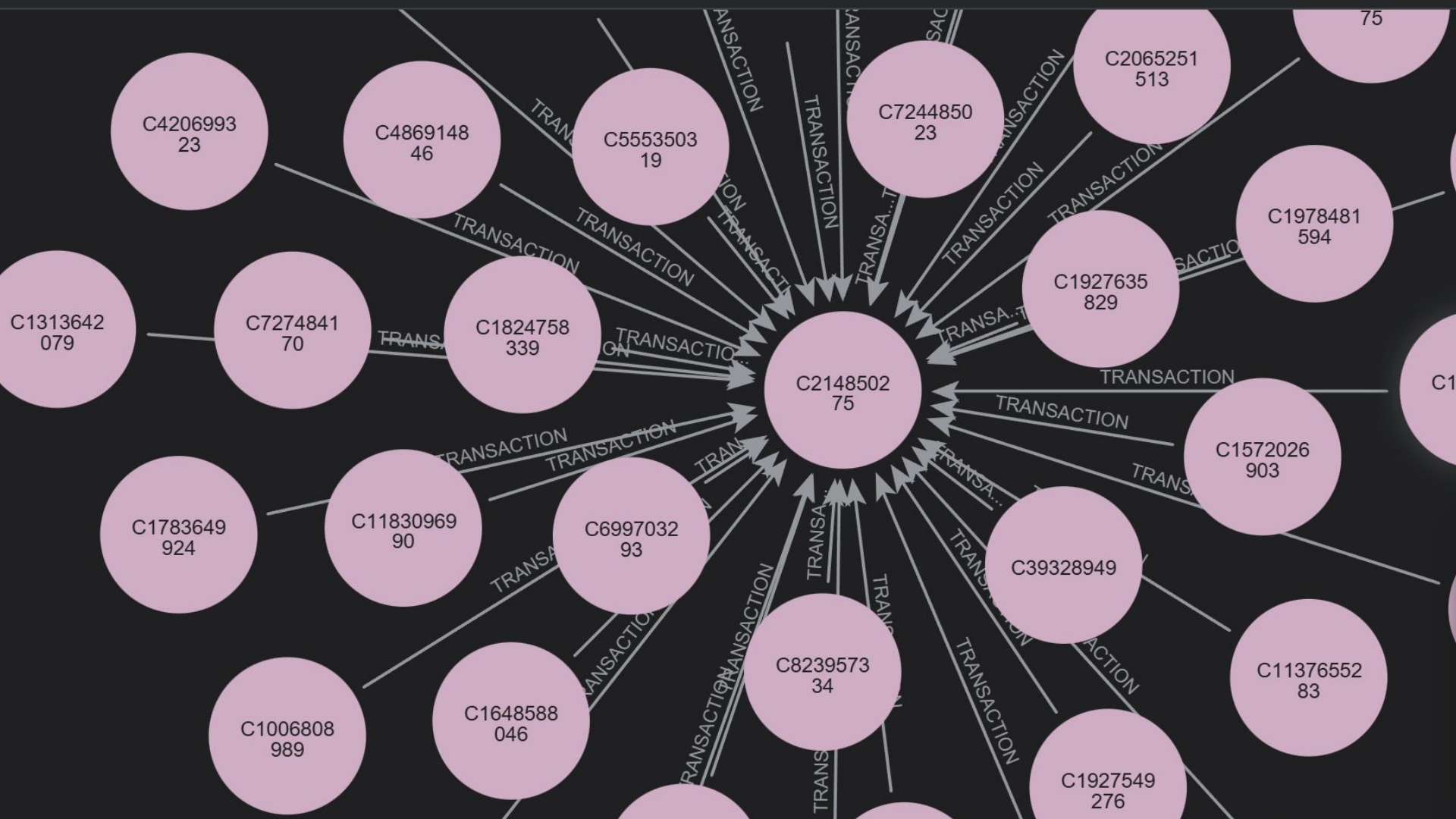
C7379234 23
C1669259 232
C1265765 939
C1856701 439
TRANSACTION
TRANSACTION
TRANSACTION
TRANSACTION


C1591531 797
C1512215 752
C1281898 359
C1220501 031
C1711643358
TRANSACTION
TRANSACTION
TRANSACTION
TRANSACTION

Case Study #1: The Low And Slow Ring

- Community ID: 9069970
- 39 members conducted 40 transactions over 706 steps (steps 9–715), suggesting 9069970 operated more patiently and slowly to avoid detection.
- $27.1M in total internal volume
- Average transaction of $677,959 with a peak single transaction of $7.69M
- Used 3 distinct transaction types to move money around the ring

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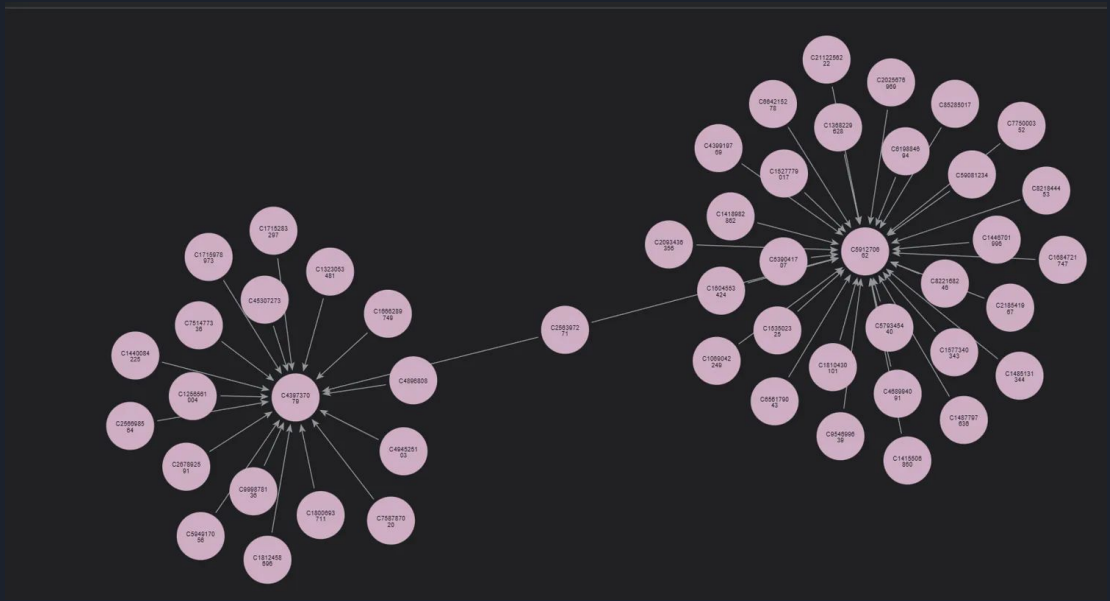
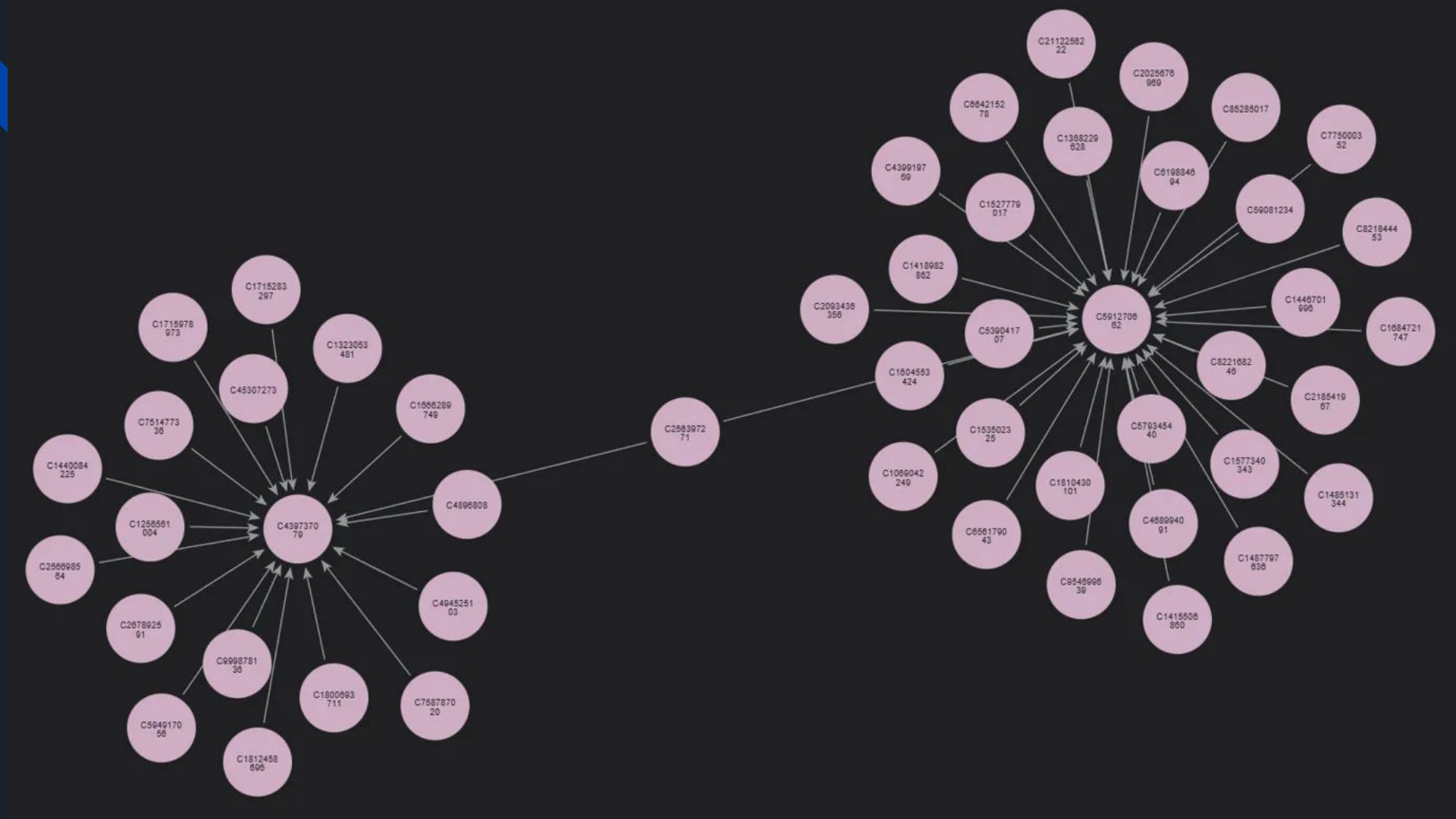
C7274841 70
C1824758 339
C1572026 903
C1783649 924
C1183096990
C699703293
C39328949
C1137655283
C1006808 989
C1648588 046
C823957334
C1927549 276
TRANSACTION

Case Study #2: The Kingpin Ring

Key facts:

- 48 members conducted 49 transactions over 388 steps (steps 16–404)
- $367M in total internal volume — by far the largest ring in the dataset
- Average transaction of $7.49M with a peak single transaction of $92.4M
- The two-hub structure suggests deliberate layering — splitting collection across two centers to avoid detection
- Used 3 distinct transaction types to obscure the money trail
- Nearly one-to-one ratio of members to transactions suggests each peripheral account transacted exactly once — classic disposable mule behavior

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C1323063 481
C45307273
C1666289 749
C7514773 36
C1440084 225
C1256561 004
C4397370 79
C4896808
C2566985 64
C4945251 03
C2678925 91
C9998781 36
C1800693 711
C7587870 20
C5949170 56
C1812458 696
C2563972 71
C21122562 22
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C8642152 78
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C1604553 424
C8221682 46
C2185419 67
C1535023 25
C5793454 40
C1069042 249
C1577340 343
C1810430 101
C1485131 344
C6561790 43
C4689940 91
C1487797 636
C9546996 39
C1415506 860

Conclusion & Future Work

Summary: Successfully narrowed 6 million transactions down to 14 high-priority targets.

Future Scope:

1. Implement Real-time flagging using Spark Streaming.
2. Use Machine Learning to auto-adjust Risk Score weights based on confirmed fraud cases.

Trend analysis

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Client Implications:

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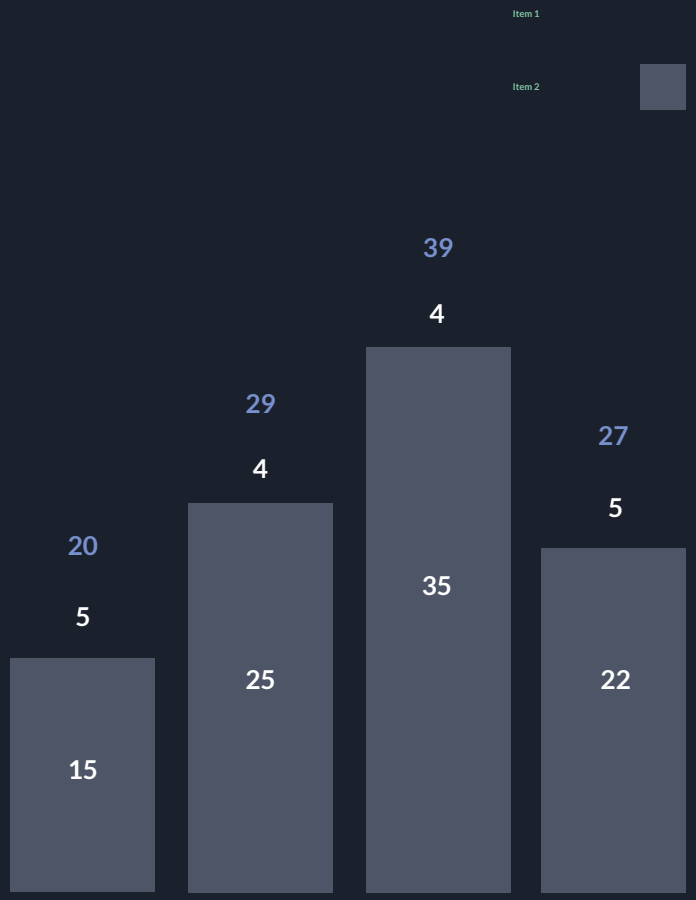
- Consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore



Proposed deliverables

Deliverable 1

- Lorem ipsum dolor sit amet
- Sed do eiusmod tempor incididunt ut labore

Deliverable 2

- Lorem ipsum dolor sit amet
- Sed do eiusmod tempor incididunt ut labore

Deliverable 3

- Lorem ipsum dolor sit amet
- Sed do eiusmod tempor incididunt ut labore

Deliverable 4

- Lorem ipsum dolor sit amet
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